Advanced Mathematics Assistant

- **Project Name:** Advanced Mathematics Assistant Platform
- **Tagline:** AI-powered math tutor with step-by-step solutions & symbolic compute
- **Team Name:** Tulip
- **Event:** HiDevs Community Hackathon
- **Date**: 06/04/2026
- **Domain: AI / EdTech**

Problem Statement

Students struggle with complex mathematics

- Traditional textbooks lack interactive, step-by-step explanations tailored to the learner
- Solving calculus, algebra, eigenvalues, probability manually is error-prone and time-consuming
- Existing tools like Wolfram Alpha are powerful but lack natural language interaction
- No unified platform combining LLM reasoning + symbolic compute + graph visualization
- Students cannot upload textbooks or scan handwritten problems for instant help

Target Audience

Who benefits from this platform?

- High school students looking for instant, step-by-step help on any math topic
- Anyone with general math questions — from basic arithmetic to calculus and statistics
- Teachers and tutors who need clear, explained solutions to share with students
- Self-learners curious about math topics and wanting AI-guided explanations
- Students who struggle with textbooks and need an interactive, conversational math tutor

Existing Solutions

Existing Solutions & Their Limitations

- Wolfram Alpha — powerful symbolic engine but lacks conversational AI and step-by-step pedagogy
- Photomath — limited to basic algebra/arithmetic, no advanced calculus or eigenvalues
- Symbolab — good for derivatives/integrals but no RAG-backed knowledge base or PDF upload
- ChatGPT / Gemini — general LLMs with no dedicated math knowledge base or graph tools
- Desmos — excellent grapher but no AI reasoning or problem-solving capabilities
- None combine: LLM reasoning + symbolic compute + RAG + graph plotting in one platform

Gaps & Opportunities

Gaps & Opportunities We Address

- No single tool combines conversational AI + symbolic math engine + visual graph plotter
- Opportunity: RAG-powered knowledge base provides context-aware math explanations
- Gap: Students cannot scan handwritten problems or upload textbook PDFs for instant solutions
- Opportunity: OCR + vision AI allows camera-based math problem extraction
- Gap: Existing tools are siloed — no unified UX from problem input to verified answer
- Opportunity: Groq's ultra-fast inference enables real-time step-by-step reasoning

Your Solution

Our Solution — Advanced Mathematics Assistant

- AI-powered chat interface: type any math problem and get step-by-step LaTeX-rendered solutions
- Graph Plotter: visualize any mathematical function with adjustable x-range and labels
- Symbolic Compute: differentiate, integrate, simplify, and solve equations using SymPy engine
- Math Topics Explorer: browse curated explanations across Calculus, Statistics, Linear Algebra, etc.
- PDF Upload: ingest textbooks or question papers into the RAG knowledge base
- Camera Scan / OCR: capture handwritten equations and extract them for AI solving

Key Features

Key Features

- Step-by-step solver with LaTeX rendering — quadratic, derivatives, integration, eigenvalues, probability
- Groq LLM (llama-3) integration for ultra-fast natural language math reasoning
- FAISS vector store + HuggingFace embeddings for context-aware RAG retrieval
- SymPy symbolic compute: differentiate, integrate, simplify, factor — all client-side accurate
- Matplotlib graph plotter with dark-theme, multi-function support, custom range & title
- PDF ingestion pipeline + Tesseract OCR for camera-captured handwritten problems

System Architecture

System Architecture

- Frontend: Streamlit UI — chat interface, sidebar tools (Graph, Symbolic, Topics, PDF, Camera)
- LLM Layer: Groq API (llama-3) via LangChain — processes user queries with math context
- RAG Pipeline: LangChain retriever → FAISS vector store → HuggingFace sentence-transformers
- Knowledge Base Builder: PDF loader + text splitter → embeddings → FAISS index stored on disk
- Symbolic Engine: SymPy — handles differentiation, integration, solving independently of LLM
- OCR Module: Pillow + Tesseract (pytesseract) → extracted text fed to LLM solver

Tech Stack

Tech Stack

- Language: Python 3.11 | Framework: Streamlit 1.32+
- LLM: Groq API (llama-3) via langchain-groq | Orchestration: LangChain 0.2+
- Vector DB: FAISS (faiss-cpu) | Embeddings: HuggingFace sentence-transformers
- Symbolic Math: SymPy 1.12 | Numerical: NumPy, SciPy | Visualization: Matplotlib
- OCR: Tesseract (pytesseract) + Pillow | PDF Parsing: PyPDF + Unstructured
- Config: python-dotenv | Storage: MongoDB (motor) for session memory

Workflow

Workflow — How It Works

- Step 1: User enters a math problem in natural language or uploads PDF / camera image
- Step 2: Query is embedded and retrieved against FAISS knowledge base for relevant context
- Step 3: Groq LLM receives question + retrieved context → generates step-by-step solution
- Step 4: SymPy engine independently verifies symbolic answers (derivatives, integrals, roots)
- Step 5: Response is rendered in the UI with LaTeX math formatting and answer highlights
- Step 6: User can visualize functions via Graph Plotter or explore related topics in sidebar

Impact

Impact & Results

- Reduces math problem-solving time from minutes to seconds with instant AI explanations
- Provides pedagogically sound step-by-step breakdowns improving student comprehension
- Supports 8+ math domains: Quadratic, Derivatives, Integration, Eigenvalues, Series, Vectors, Statistics, Probability
- Enables self-paced learning — available 24/7 without needing a human tutor
- RAG-backed answers grounded in textbook knowledge, reducing hallucinations
- Empowers educators to quickly generate verified solutions for teaching material

Future Scope

Future Scope

- Multi-modal input: handwriting recognition via advanced vision models (GPT-4o / Claude Vision)
- Collaborative whiteboard — share and solve problems with peers in real-time
- Personalized learning paths — adaptive difficulty based on user history and performance
- Voice interface — speak math problems aloud, receive spoken step-by-step solutions
- Mobile app (React Native) for on-the-go problem solving and camera scanning
- Gamified practice mode — timed challenges, leaderboards, achievement badges

Team Info

Prashant J : AI/ML Engineer — Groq API, RAG pipeline
Mikhil Sai: Backend Developer — FAISS, embeddings
Chandan Kumar: Frontend Developer — Streamlit UI, graphs
Sumadha Chhetri: Math Expert — SymPy engine, OCR, topics

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New: Upload PDF textbooks or question papers!

Use the sidebar for graphs, symbolic compute, PDF upload and camera scan.

Groq API Connected

Solve $2x^2 + 5x - 3 = 0$ step by step

Question: Solve $2x^2 + 5x - 3 = 0$ step by step

Step 1 — Apply the Quadratic Formula To solve the quadratic equation $2x^2 + 5x - 3 = 0$, we will use the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here, $a = 2, b = 5$, and $c = -3$. Substituting these values into the formula gives us $x = \frac{-5 \pm \sqrt{5^2 - 4*2*(-3)}}{2*2}$.

Step 2 — Simplify the Expression We simplify inside the square root first: $5^2 - 4 * 2 * (-3) = 25 + 24 = 49$. So, the equation becomes $x = \frac{-5 \pm \sqrt{49}}{4}$. Since $\sqrt{49} = 7$, we have $x = \frac{-5 \pm 7}{4}$.

Step 3 — Solve for Both Values of x We solve for both values of x: $x_1 = \frac{-5+7}{4} = \frac{2}{4} = \frac{1}{2}$ and $x_2 = \frac{-5-7}{4} = \frac{-12}{4} = -3$.

Answer: $x = \frac{1}{2}$ or $x = -3$

View Context Sources

Copy plain text

Download Answer as Markdown

Tips

Quick Examples

- Quadratic
- Derivatives
- Integration
- Eigenvalues
- Statistics
- Series
- Vectors
- Probability

e.g., Solve $x^2 - 5x + 6 = 0$ or Find the derivative of $\sin(x) \cdot x^2$

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Graph Plotter

Function(s):

x**2

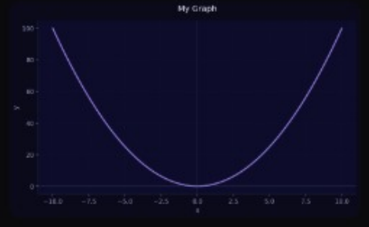
x range

-10 10

Title:

My Graph

Plot



Symbolic Compute

Expression:

x**3 + 2*x

Action:

Differentiate

Step 3 — Solve for Both Values of x We solve for both values of x: $x_1 = \frac{-5+7}{4} = \frac{2}{4} = \frac{1}{2}$ and $x_2 = \frac{-5-7}{4} = \frac{-12}{4} = -3$.

✅ Answer: $x = \frac{1}{2}$ or $x = -3$

View Context Sources

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Download Answer as Markdown

Find the derivative of f(x) = x^3sin(x)

Question: Find the derivative of f(x) = x^3sin(x)

Step 1 — Apply the Product Rule To find the derivative of $f(x) = x^3 \sin(x)$, we use the product rule, which states that if $f(x) = u(x)v(x)$, then $f'(x) = u'(x)v(x) + u(x)v'(x)$. Here, $u(x) = x^3$ and $v(x) = \sin(x)$. So, $u'(x) = 3x^2$ and $v'(x) = \cos(x)$.

Step 2 — Calculate the Derivative Applying the product rule, we get $f'(x) = (3x^2)\sin(x) + x^3\cos(x)$.

✅ Answer: $f'(x) = 3x^2 \sin(x) + x^3 \cos(x)$

Copy plain text

Download Answer as Markdown

Tips

e.g., Solve $x^2 - 5x + 6 = 0$ or Find the derivative of $\sin(x) \cdot x^2$

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Sym

Expression

x**2+3=0

Action:

Differentiate

Compute

Math Topics Explorer

Select a topic to get a comprehensive explanation

Category:

Statistics

Topic:

Probability Distributions

Explain Topic

Upload PDF

Upload textbook, notes or question paper

Upload

Question: Provide a comprehensive and detailed explanation of Probability Distributions in Statistics

Introduction to Probability Distributions

Probability distributions are a fundamental concept in statistics, describing the probability of different values or ranges of values that a random variable can take. They are essential for modeling and analyzing real-world phenomena, making predictions, and estimating uncertainties.

Key Concepts

- **Random Variable**: A variable whose possible values are determined by chance events.
- **Probability Density Function (PDF)**: A function that describes the probability of a continuous random variable taking on a particular value.
- **Cumulative Distribution Function (CDF)**: A function that gives the probability that a random variable takes on a value less than or equal to a given value.
- **Expected Value**: The long-run average value of a random variable.
- **Variance**: A measure of the spread or dispersion of a random variable.

Common Probability Distributions

1. **Normal Distribution (Gaussian Distribution)**
   - PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
   - CDF: $\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dt$
   - Mean: μ
   - Variance: σ^2
2. **Binomial Distribution**
   - PMF: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
   - Mean: np
   - Variance: $np(1-p)$
3. **Poisson Distribution**
   - PMF: $P(X = k) = \frac{\lambda^k e^{-\lambda}}{}$

e.g., Solve $x^2 - 5x + 6 = 0$ or Find the derivative of $\sin(x) \cdot x^2$